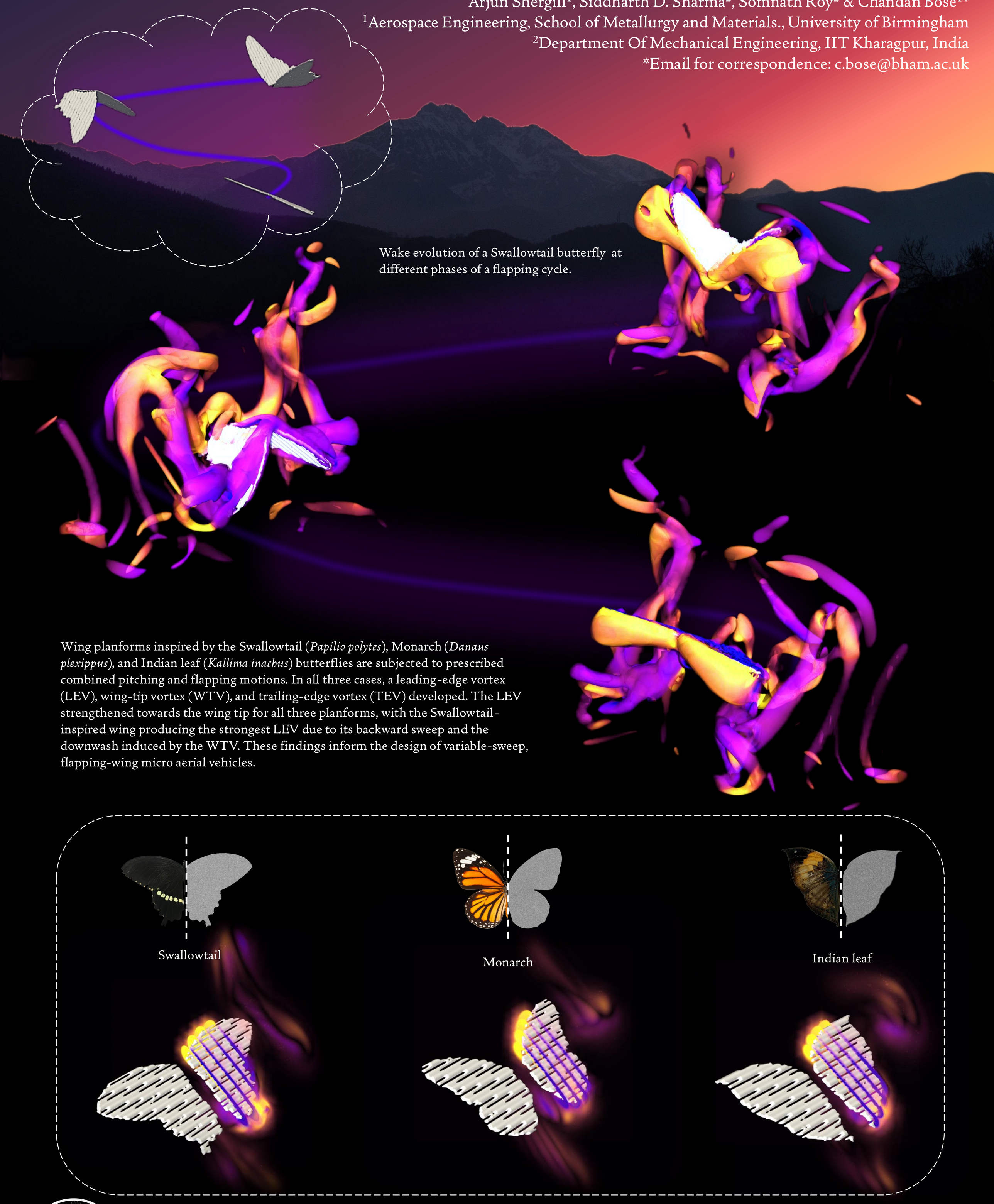
Wing-Wake Interactions across Different Butterfly Species

Arjun Shergill[I], Siddharth D. Sharma[2], Somnath Roy[2] & Chandan Bose[I]*
[I]Aerospace Engineering, School of Metallurgy and Materials., University of Birmingham
[2]Department Of Mechanical Engineering, IIT Kharagpur, India
*Email for correspondence: c.bose@bham.ac.uk



Wake evolution of a Swallowtail butterfly at different phases of a flapping cycle.

Wing planforms inspired by the Swallowtail (*Papilio polytes*), Monarch (*Danaus plexippus*), and Indian leaf (*Kallima inachus*) butterflies are subjected to prescribed combined pitching and flapping motions. In all three cases, a leading-edge vortex (LEV), wing-tip vortex (WTV), and trailing-edge vortex (TEV) developed. The LEV strengthened towards the wing tip for all three planforms, with the Swallowtail-inspired wing producing the strongest LEV due to its backward sweep and the downwash induced by the WTV. These findings inform the design of variable-sweep, flapping-wing micro aerial vehicles.

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